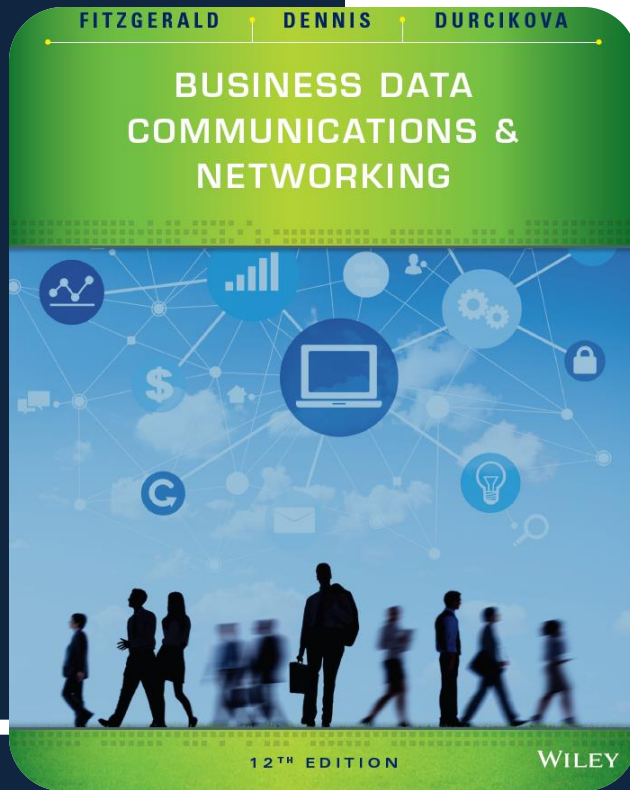




BUSINESS DATA COMMUNICATIONS & NETWORKING

Chapter 3 Physical Layer

FitzGerald • Dennis • Durcikova

Outline

- Circuits and Data Flow
- Multiplexing
- Media
- Digital Transmission of Digital Data
- Analog Transmission of Digital Data
- Digital Transmission of Analog Data
- Implications for Management

Physical Layer

- Layer 1 in the Internet model
- Focus on transmission over circuits
- Types of Circuits
 - **Physical circuits** connect devices & include wires
 - **Logical circuits** refer to the transmission characteristics of the circuit

Internet Model

Application

Transport

Network

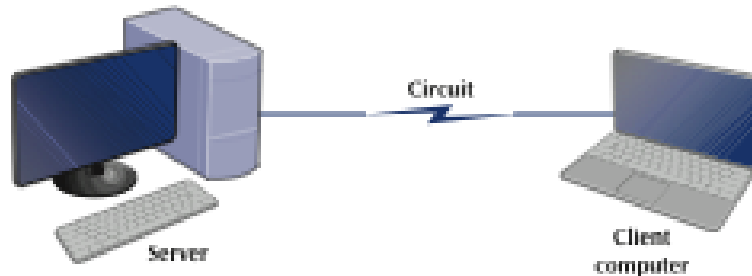
Data Link

Physical

Circuits

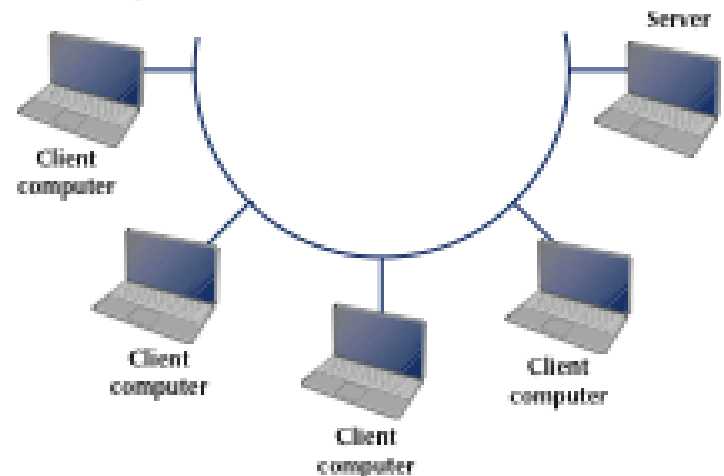
- Circuit Configuration
 - **Point-to-Point** (**dedicated**) circuits include most wired connections today
 - **Multipoint** (**shared**) circuits are most commonly used in wireless today. Shared circuits are less expensive

FIGURE 3-1
Point-to-point circuit



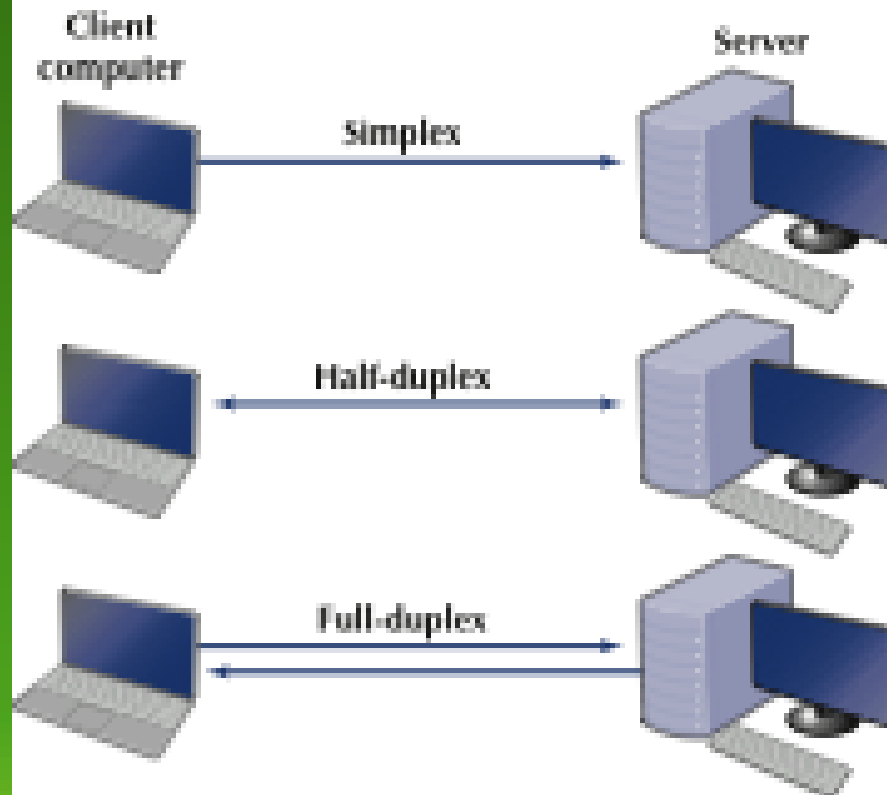
one computer to another computer

FIGURE 3-2
Multipoint circuit



Data Flow

FIGURE 3-3
Simplex, half-duplex,
and full-duplex
transmissions

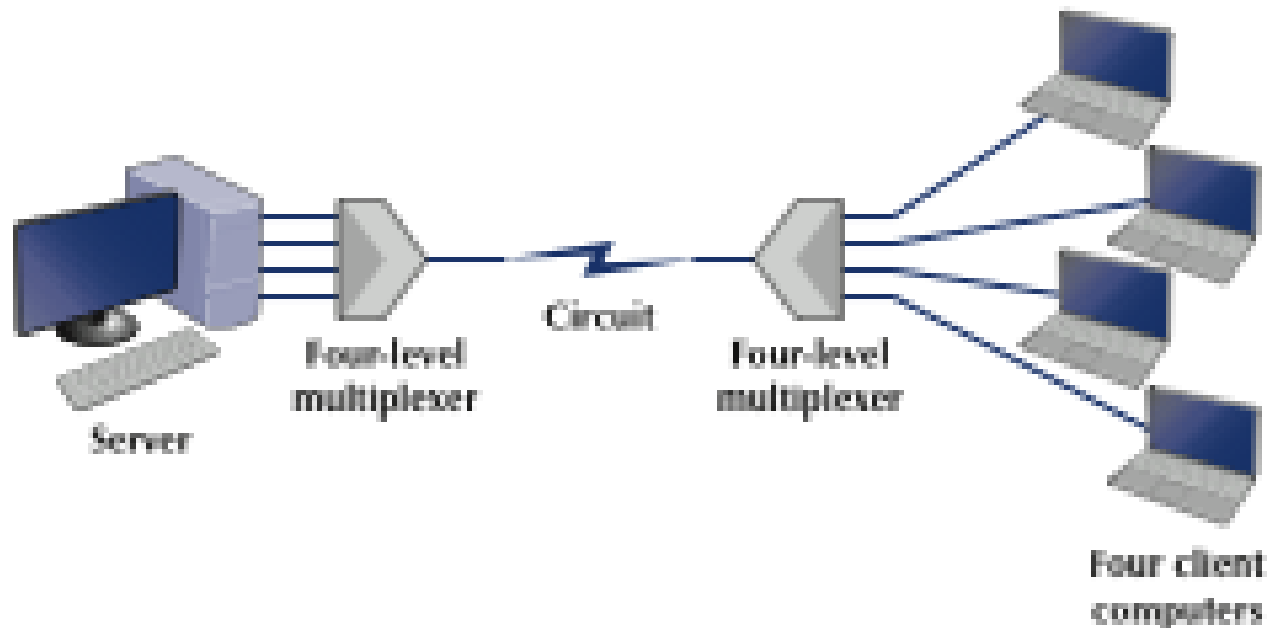


- Data flows in **one direction** (radio or TV transmission)
- Data flows **both directions**, but only **one at a time** (walkie-talkie link)
- Data flows simultaneously in **both directions**, with no turnaround time

Multiplexing

- Divide high-speed circuit into several slower circuits
- Main advantage is **cost**
- Categories of multiplexing
 - Frequency/Wavelength
 - Time

FIGURE 3-4
Multiplexed circuit



Multiplexing

- **Frequency Division Multiplexing (FDM)**
 - Used for analog transmission
 - Creates “channels” from larger frequency band
 - **Guardbands** separate channels to prevent interference



Unused part of the radio spectrum.

Multiplexing

- **Wavelength Division Multiplexing (WDM)**
 - A variant of FDM used in **fiber optic** circuits
 - Makes use of **multiple light wavelengths** (colors) to divide circuit into channels
 - Dense WDM can divide circuit into more than **100 channels** per fiber each transmitting at **10 Gbps**

Multiplexing

- **Time Division Multiplexing (TDM)**
 - Used for digital transmission
 - Circuit is divided by devices **taking turns**
 - In traditional TDM, all have **equal** turns
 - More **efficient than FDM**
 - **No guardbands**, but may have idle time slots
- **Statistical Time Division Multiplexing (STDM)**
 - A variation of TDM
 - Designed to reduce idle time slots by **allocating slots based on statistical network usage**
 - Disadvantages
 - **Potential time delays** when actual usage does not match statistically allocated time slots
 - **Additional logical addressing** requirements

Multiplexing

- **Inverse multiplexing**
 - Combines many low-speed circuits into one high-speed circuit
 - e.g., two T-1 lines multiplexed (creating a capacity of $2 \times 1.544\text{Mbps} = 3.088\text{ Mbps}$)

Media

Physical matter used to carry voice or data transmissions

- **Guided media** – transmission flows along physical environment (twisted pair wire, coaxial cable, or fiber optic cable)
- **Wireless (Radiated) media** - transmission flows through the air

Guided Media

- Twisted-pair (TP) cable
 - Insulated pairs of wires bundled together
 - Wires twisted to reduce electromagnetic interference
 - Some times use additional shielding (STP)
 - Commonly used for telephones, LANs
 - Characteristics
 - Price – inexpensive
 - Distance – typically up to 100m
 - Use - Telephones, LANs

FIGURE 3-5

Category 5e twisted pair wire

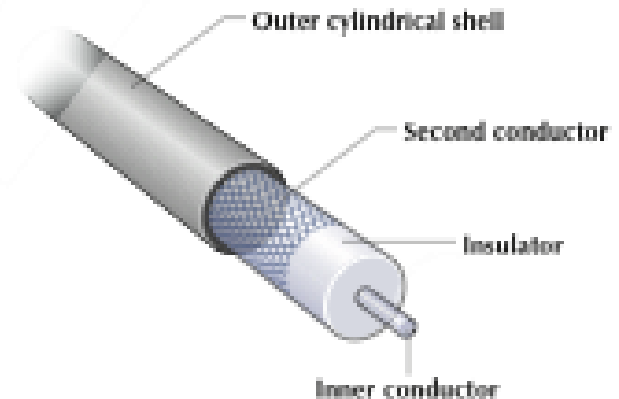
Source: Courtesy of Belkin International, Inc.



Guided Media

- Coax cable
 - Has a single copper core, plus outer insulation, shielding, and inner insulation
 - Less prone to interference
 - Characteristics
 - Price - inexpensive (but more costly than TP)
 - Distance - up to 2 km (1.2 miles)
 - Use: Cable TV / Internet

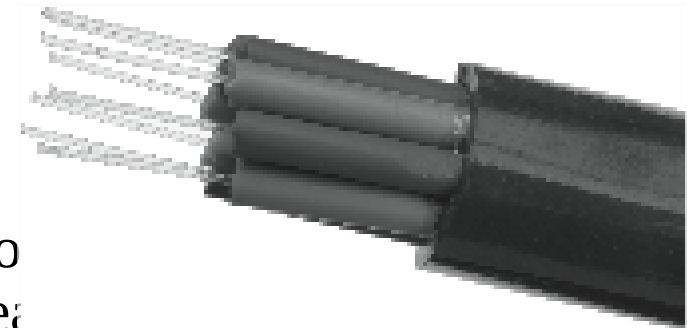
Source: Courtesy of Tim Kloske



Guided Media

- Fiber optic cable
 - Optical core made of glass or plastic
 - Data transmitted using light from lasers or LEDs
 - Resistant to interference and corrosion
 - Extremely fast data rates
 - Characteristics
 - Price: Expensive
 - Distance: 500m – 100km
 - Use: Trunk line / Backbone, long distance circuits (e.g., undersea cables)

Source: © Hugh Threlfall/Alamy



Guided Media

- Fiber optics
 - Multimode (about 50 micron core)
 - Graded index multimode
 - Single mode (about 5 micron core)

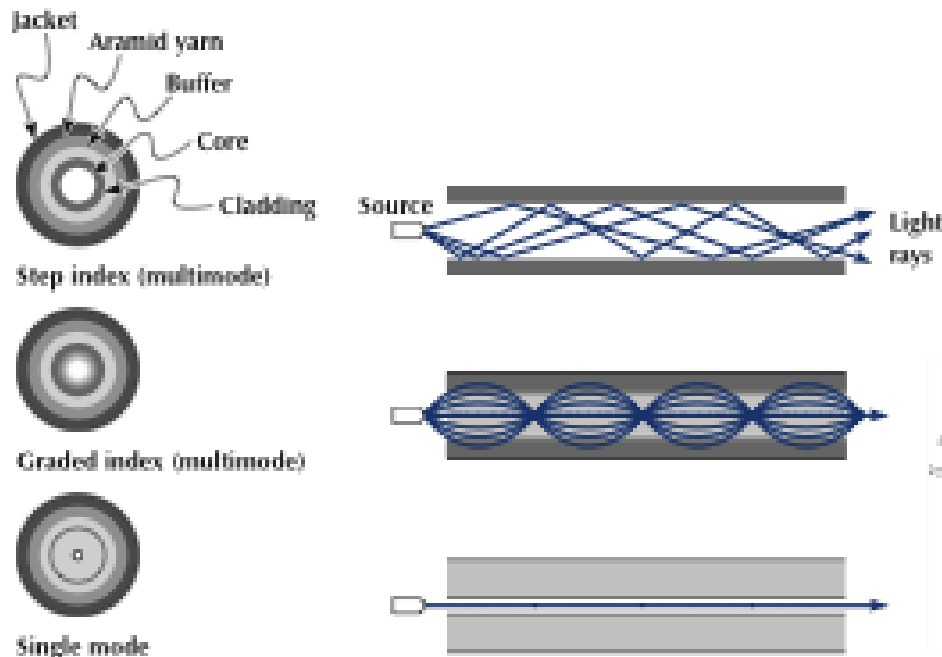


FIGURE 3-7 Fiber-optic cable

Source: © Hugh Threlfall/Alamy

Wireless Media

- Radio
 - Wireless transmission of electrical waves through air
 - Each device on network has a radio transceiver operating at a **specific frequency range**
 - Enables mobile network communication
 - Characteristics
 - Distance: depends on frequency and power
 - Use: Wireless LANs, cellular and cordless phones, baby monitors

Wireless Media

- Infrared
 - Low frequency light waves (below the visible spectrum)
 - Similar to technology used in infrared TV remote controls.
 - Prone to interference, particularly from heavy rain, smoke, and fog that obscure the light transmission
 - Not very common
 - Transmit data from building to building

Wireless Media

- Microwave
 - High-frequency radio communication
 - Requires line of sight which may require large antennas and towers
 - Affected by weather
 - Characteristics
 - Distance: ~60 km (due to curvature of earth)
 - Use: Trunk line / Backbone, long distance
- Satellite
 - Special form of microwave communication
 - Long distance leads to propagation delays

FIGURE 3-8

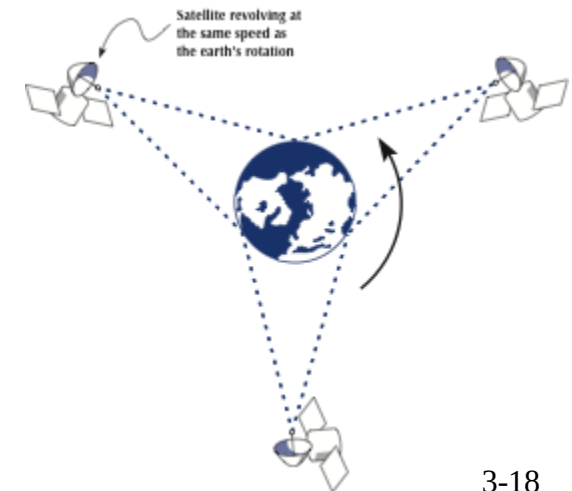
A microwave tower. The round antennas are microwave antennas and the straight antennas are cell phone antennas

Source: © Matej Pribelsky / iStockphoto



FIGURE 3-9

Satellites in operation

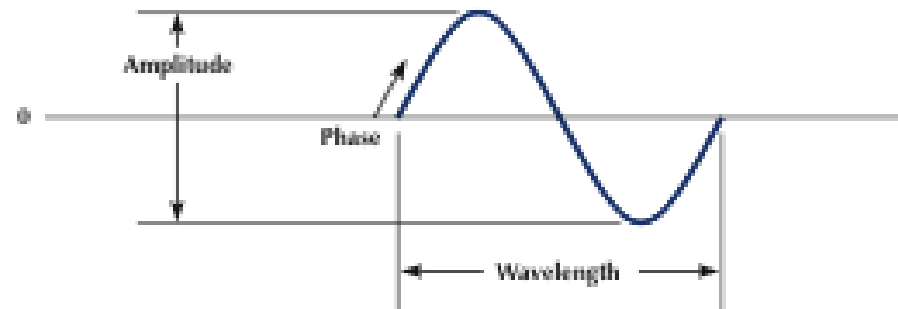
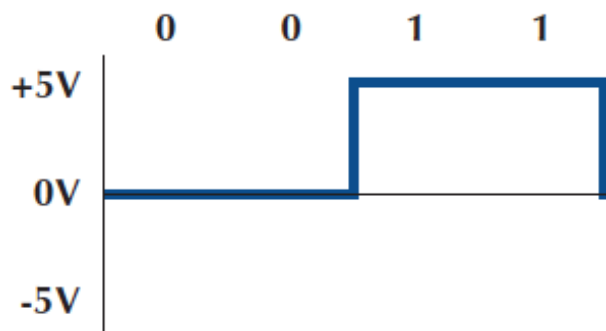


Media

- Factors to consider in media selection
 - Type of network
 - Cost
 - Transmission distance
 - Security
 - Error rates
 - Transmission speeds

Digital vs. Analog Data

- **Digital** transmission involves discrete binary values (i.e., 0 or 1)
- **Analog** transmission involves continuous waves



Digital Transmission of Digital Data

- Coding scheme needed to ensure sender and receiver understand messages (e.g., ASCII, Unicode, etc.)
- Coding : Representation of one set of symbols by another set of symbols
- A character is represented by a group of bits

FIGURE 3-10

Binary numbers used to represent different characters using ASCII

Character	ASCII
A	01000001
B	01000010
C	01000011
D	01000100
E	01000101
a	01100001
b	01100010
c	01100011
d	01100100
e	01100101
1	00110001
2	00110010
3	00110011
4	00110100
!	00100001
\$	00100100

Digital Transmission of Digital Data

- Transmission modes

1. **Parallel:** multiple bits transmitted simultaneously

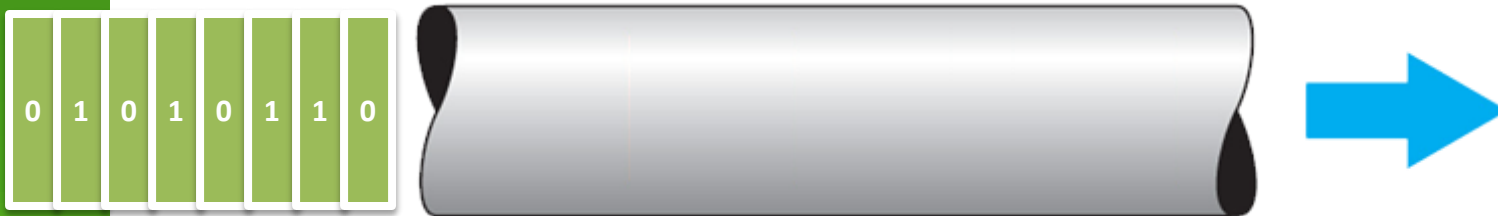
The way the internal transfer of binary data takes place inside a computer. If the internal structure of the computer is 8-bit, then all eight bits of the data element are transferred between main memory and the central processing unit simultaneously on 8 separate connections.



Each character is followed by a time delay

Digital Transmission of Digital Data

- Transmission modes
 2. **Serial:** bits are transferred sequentially, one at a time. Each bit followed by a time delay



Digital Transmission of Digital Data

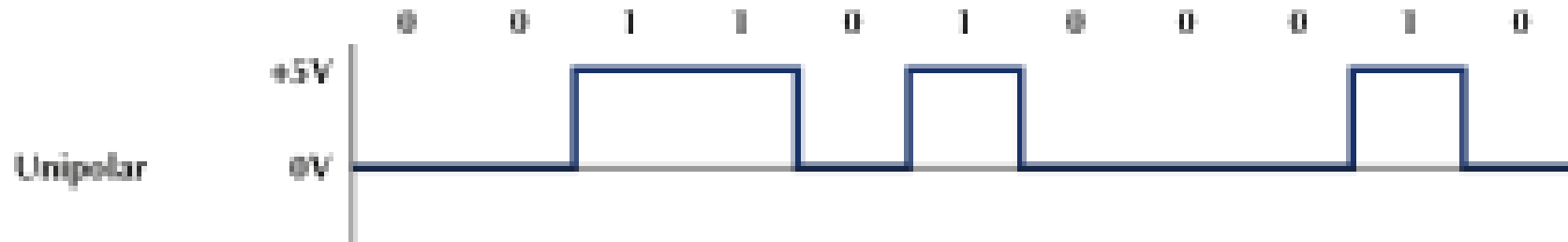
- Sender and receiver must agree upon:
 - **Set of symbols**
 - How bits are encoded as voltages or light pulses
 - e.g., +5V might be encoded as a “1”
 - **Symbol rate**
 - How often symbols are sent
 - e.g., with a symbol rate of 64 kilohertz (kHz), a symbol is sent every 1/64,000 of a second

Digital Transmission of Digital Data

- Five types of signaling techniques
 1. **Unipolar** - voltage is 0 or positive representing binary bits (in some circuits, 0 and negative voltage could be used)

FIGURE 3-13

Unipolar, bipolar, and Manchester signals (digital)

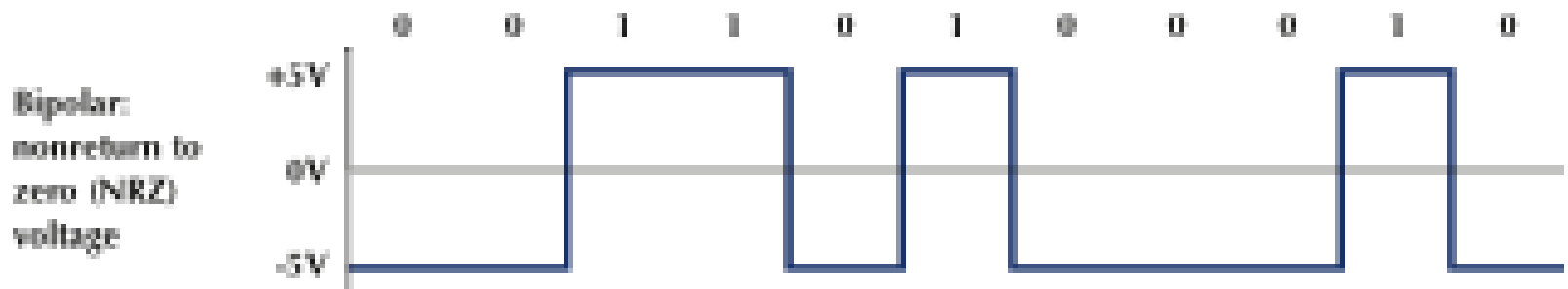


Digital Transmission of Digital Data

- Five types of signaling techniques
 2. **Bipolar NRZ (nonreturn to zero)** - voltage is positive or negative, but not zero
 - Less errors than unipolar because signals are more distinct

FIGURE 3-13

Unipolar, bipolar, and Manchester signals (digital)

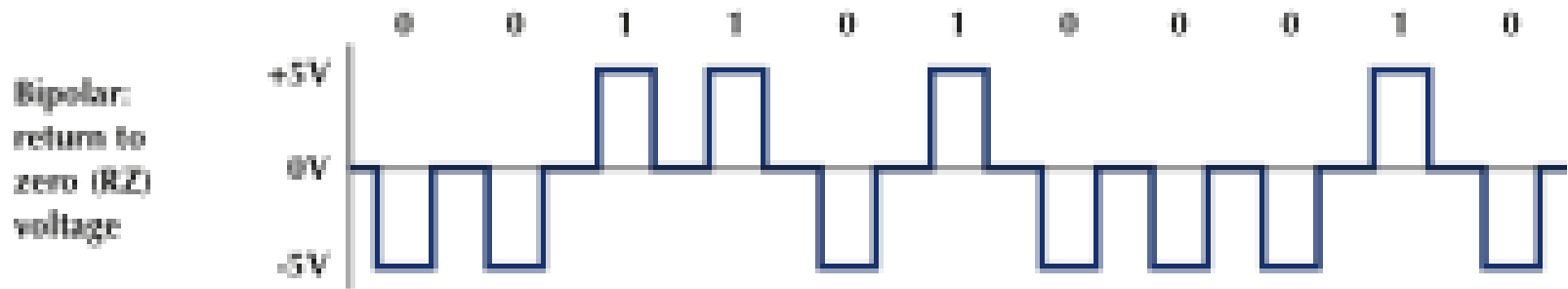


Digital Transmission of Digital Data

- Five types of signaling techniques
 3. **Bipolar RZ (return to zero)** - voltage is positive or negative, returning to zero between each bit
 - Less synchronization errors than bipolar NRZ

FIGURE 3-13

Unipolar, bipolar, and Manchester signals (digital)

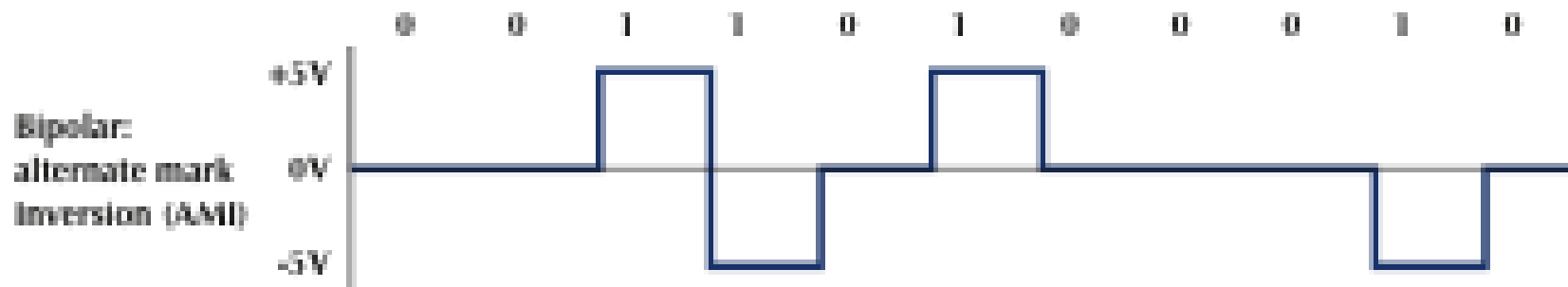


Digital Transmission of Digital Data

- Five types of signaling techniques
 4. **Bipolar AMI (Alternate mark inversion)** - voltage is zero, positive, or negative, returns to zero between each bit, and **alternates** between positive and negative voltage

FIGURE 3-13

Unipolar, bipolar, and Manchester signals (digital)

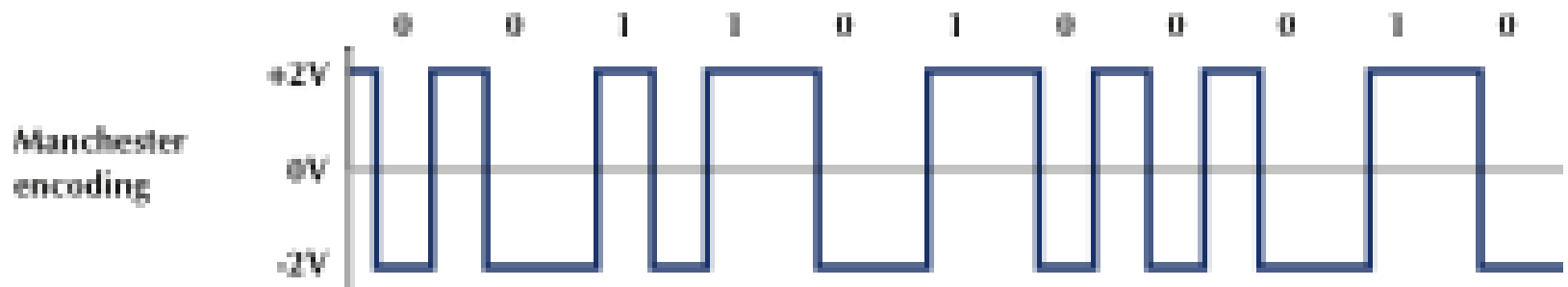


Digital Transmission of Digital Data

- Five types of signaling techniques
 5. **Manchester** (special type of Bipolar NRZ)
 - Voltage is positive or negative and bits are indicated by a mid-bit transition
 - More popular

FIGURE 3-13

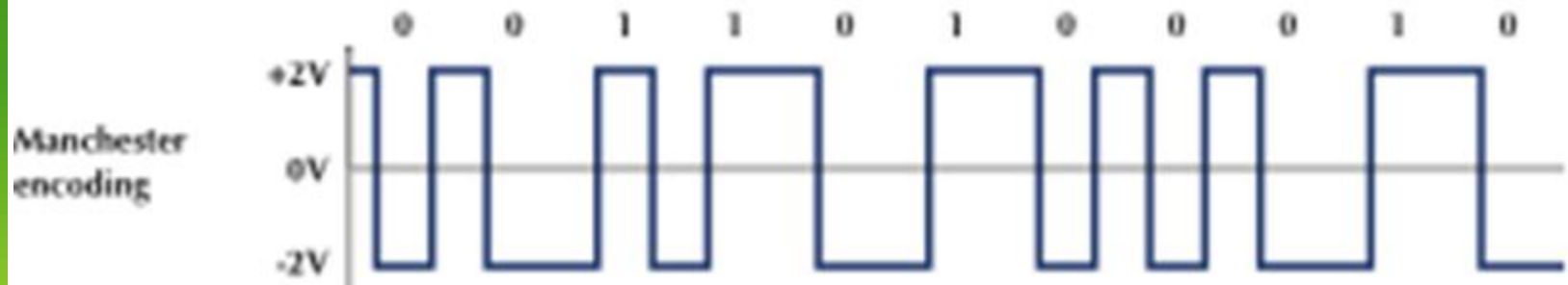
Unipolar, bipolar, and
Manchester signals
(digital)



Digital Transmission of Digital Data

In Manchester encoding,

- The signal is changed from a high to low or from low to high in the middle of the signal.
- A change from high to low is used to represent a 1 (or a 0), while the opposite (a change low to high) is used to represent a 0 (or a 1).
- **Less susceptible to having errors go undetected**, because if there is no transition in mid-signal the receiver knows that an error must have occurred. (There should be transition **always**)

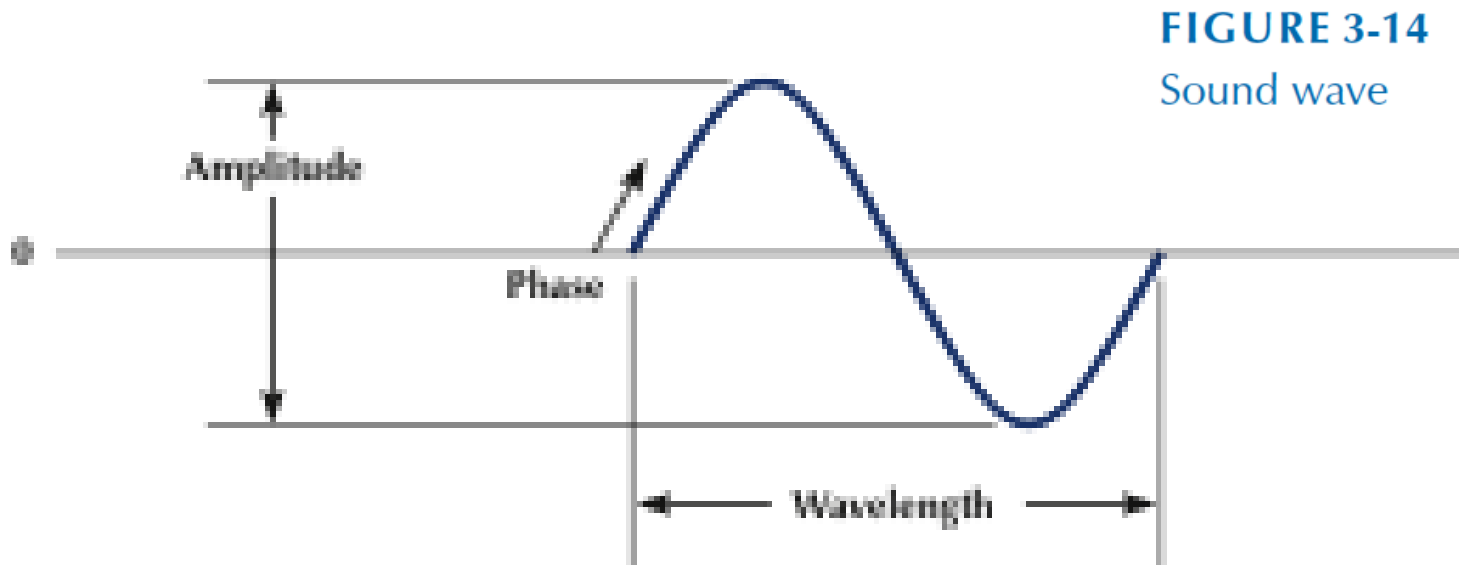


Analog Transmission of Digital Data

- Telephone system built for analog data
 - Electrical signals mimic sound waves (i.e., voice)
 - Analog transmissions take on range of values (vs. discrete values of digital transmissions)
 - Need a **modem** (**mo**dulator/**dem**odulator) to **convert** from analog to digital and vice versa

Analog Transmission of Digital Data

- Three characteristics of waves
 1. **Amplitude:** height of wave (decibels)
 2. **Frequency:** waves per second (hertz)
 - **Wavelength** is the inverse of frequency
 3. **Phase:** wave direction (degrees) or the point at which the wave begins

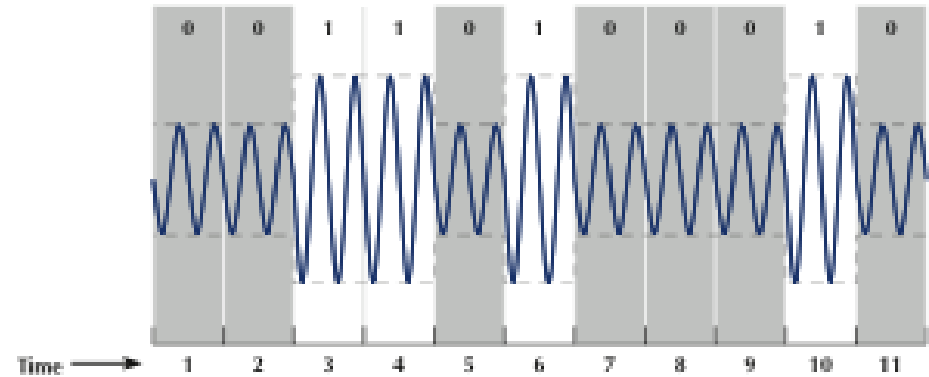


Analog Transmission of Digital Data

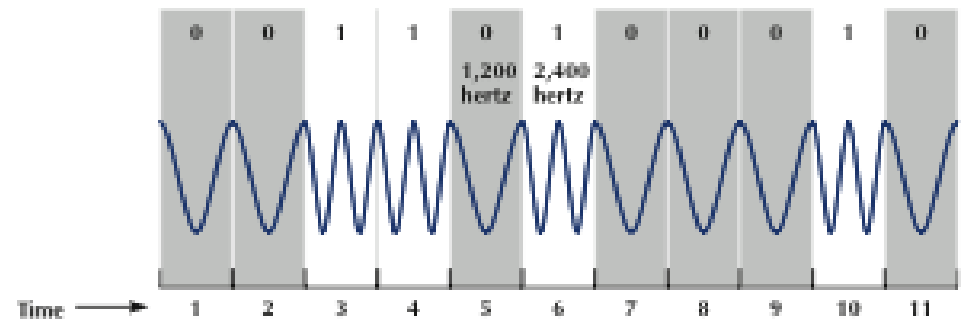
- **Carrier wave** is basic wave transmitted through a circuit
- **Modulation** is the **modification** of a carrier wave's fundamental characteristics in order to **encode information**
- Three ways to modulate a carrier wave:
 1. **Amplitude Modulation (AM)** or Amplitude Shift Keying (ASK)
 2. **Frequency Modulation (FM)** or Frequency Shift Keying (FSK)
 3. **Phase Modulation (PM)** or Phase Shift Keying (PSK)

Analog Transmission of Digital Data

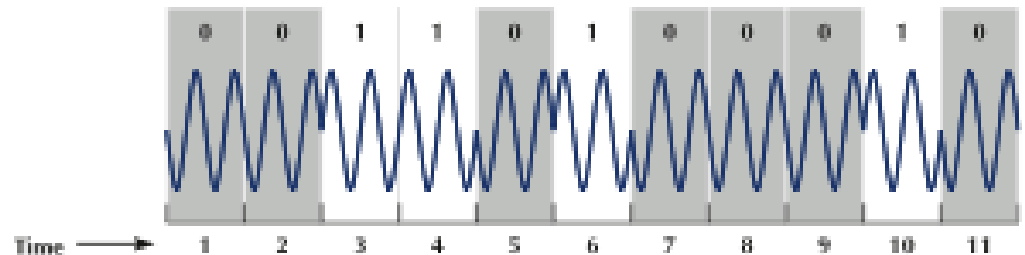
- Amplitude Modulation



- Frequency Modulation



- Phase Modulation



Analog Transmission of Digital Data

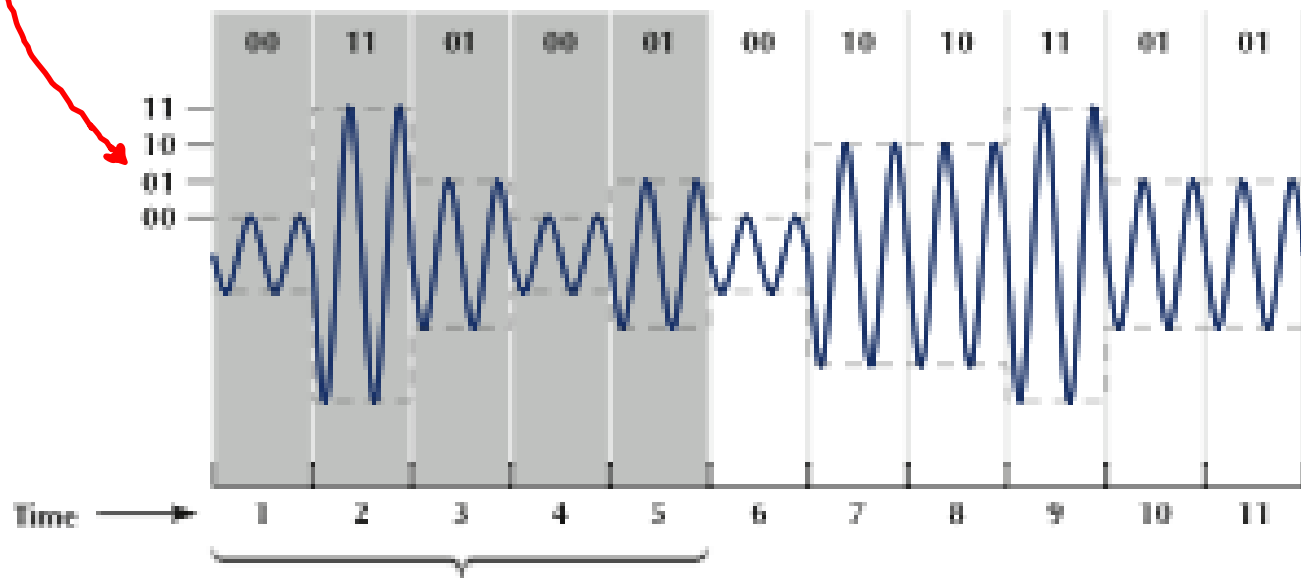
- **Symbol:** One or more modifications to a carrier wave used to encode data
- We can send 1 bit by defining two different symbols (e.g., amplitudes, frequencies, etc.)
- We can send multiple bits by defining more than two symbols
 - 1 bit of information $\rightarrow$ 2 symbols
 - 2 bits of information $\rightarrow$ 4 symbols
 - 3 bits of information $\rightarrow$ 8 symbols
 - n bits of information $\rightarrow 2^n$ symbols

Analog Transmission of Digital Data

- Two-bit Amplitude Modulation (AM)
 - With 4 levels of amplitude defined as symbols, 2 bits can be transmitted per symbol

FIGURE 3-18

Two-bit amplitude modulation



This data took 10 symbols
with 1-bit amplitude modulation

Analog Transmission of Digital Data

- Frequency modulation (FM)
 - Technique whereby each 0 or 1 is represented by a number of waves per second (i.e., a different frequency).
 - The amplitude does not vary.
 - One frequency (i.e., a certain number of waves per second) is defined to be a one, and a different frequency (a different number of waves per second) is defined to be a one.

Analog Transmission of Digital Data

- Phase modulation (PM)
 - Phase refers to the direction in which the wave begins.
 - Until now, the waves we have shown start by moving up and to the right (this is called a 0° phase wave).
 - Waves can also start down and to the right. (phase of 180°).
 - With PM, one phase is defined to be a zero and the other phase is defined to be a one.

Analog Transmission of Digital Data

- **Data rate** (bit rate): number of bits transmitted per second
- **Symbol rate**: number of symbols transmitted per second

$$\text{Data rate} = \text{symbol rate} \times (\# \text{ bits/symbol})$$

- Example

Symbol rate = 16,000 symbols/sec

#bits/symbol = 4 bits/symbol

Data rate = 16,000 symbols/sec $\times$ 4 bits/symbol
= 64,000 bits/sec = 64Kbps

Digital Transmission of Analog Data

- **Codecs (COfde, DECode)** is a device or software that converts an analog signal (e.g., voice) into digital form and the reverse
- **Pulse-Code Modulation (PAM)** converts analog to digital by:
 1. Sampling the analog signal at regular intervals
 2. Measuring the amplitude of each sample
 3. Encoding (quantizing) the amplitude as binary data (A2D)
- **Quantizing Error** is the difference between the original analog signal and the approximated digital signal
- Reducing quantizing error can be done by:
 - Sampling more frequently
 - Using more levels of amplitude in encoding

Digital Transmission of Analog Data

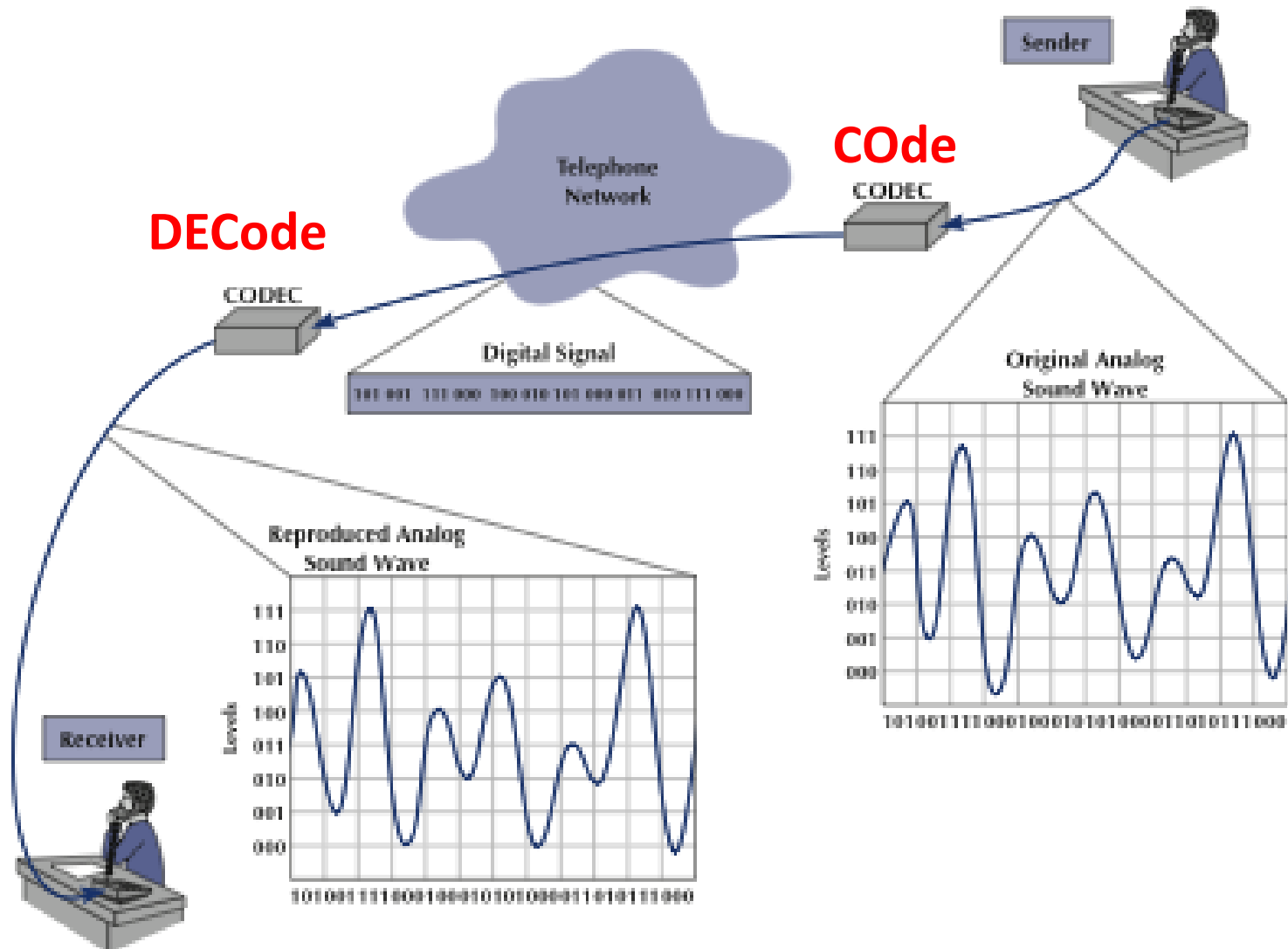


FIGURE 3-20 Pulse amplitude modulation (PAM)

Implications for Management

- Digital cabling tends to be least expensive and most reliable
- Data and voice networks continue to converge
- Wired networks remain more secure and reliable than wireless